**Individual Project : Student Hangout Creator, 2025-2026 Instruction Manual**

## Introduction

This application is intended for university students who wish to easily discover, host and join activities on or off campus. The main purpose is to encourage community and reduce social isolation by providing an easy way to organize and find spontaneous or planned events.

The primary functionalities of this application are:

- **Hosting a Hangout:** Users can create new events with details like title, location, participant limits and choose from a list of predetermined categories.
- **Joining a Hangout:** Users can browse active events and join them instantly if space is available.
- **Real-Time Dashboard:** A live feed of all events that updates as new events are created.
- **My activities:** A personalized view where users can track events they are hosting and events they have joined.

By using this application, users will benefit from an easy, safe and efficient way to improve their social lives without relying on physical bulletin boards.

## Design

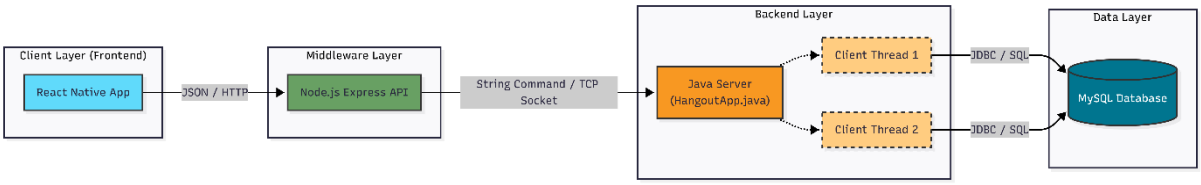
The application architecture is based on the following technologies:

- **Frontend**: React Native, Expo.
- **Middleware**: Node.js, Express.
- **Backend**: Java 17.
- **Database**: MySQL 8.0.

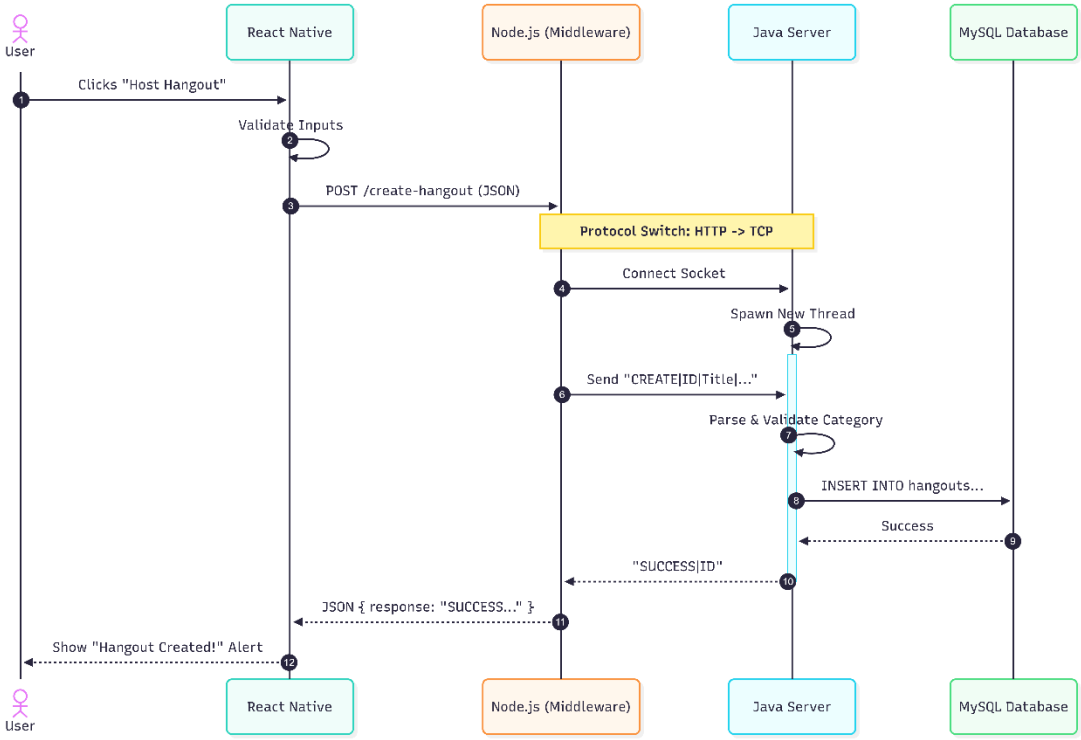
## Diagrams

Below are the architectural diagrams explaining the interactions between the components.

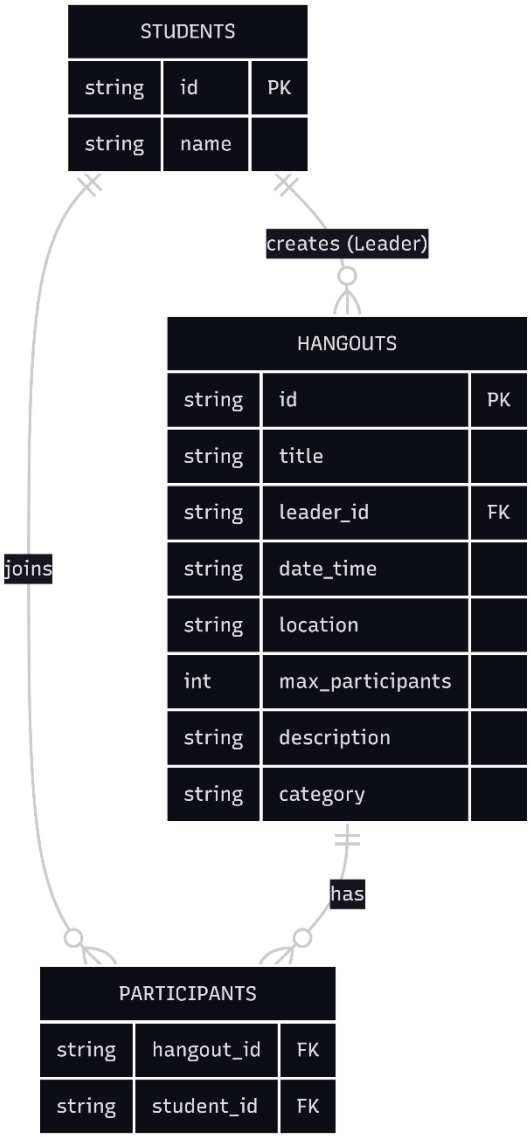
**System Architecture:** Illustrating the flow of data from the mobile client, through the API Middleware, to the Java backend and database.




**Sequence Diagram**: Showcasing the process of a user hosting a hangout.



**Entity Relationship Diagram**: Showcasing the way the database manages many-to-many relationships between students and events.




## Usage

Below are examples of how the user accesses the main functionalities.

1. **Authentication and Registration**: Upon opening the app, the user is presented with a Login screen. They enter their student ID. If the ID is new, the system automatically registers them into the database. If it exists, it logs them in.

2. **Interface Navigation**: The application features a bottom navigation bar with three main tabs:
   - Dashboard(Home): The main feed of all active hangouts.

- My activity: A split view showing events you are hosting and events you have joined.
- Profile: A user summary showing the student's name and ID.

3. **Using main functions:** The application offers several functionalities for users:
    - Hosting a Hangout: Users can tap the floating "+" button on the home screen. They fill out a form containing information about the event structure. Upon submission, the backend validates the data and creates an event in the database.

    - Joining a Hangout: Tapping on any card on the dashboard opens a "Details" modal where the user can join the respective hangout. The system prevents joining if the event is full or if the user is already a member.

    - Deleting an event: In the "My Activity" tab, users can delete events they are hosting. This action permanently deletes all information and participant data associated with that event in the database.

## Conclusions and Next Steps

This application provides a robust solution for student social coordination. Users benefit from an optimized and efficient experience, and the application can be improved by:

- **Adding chat functionality:** Implementing real-time messaging for participants withing the app.
- **Push Notifications:** Alerting users when someone joins their hangout.
- **Advanced Filtering:** Allowing users to filter the hangouts on the Dashboard by date, specific terms, categories, etc.